SAMPLE FINAL EXAMINATION

Sample

Question	Points	Score
Basic concepts	56	
SVM and Kernel Methods	9	
Neural Network	10	
PCA	10	
Overfitting	15	
Total:	100	

- Please write down your **name** and **student ID** on the **answer paper**.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

Question 1 [56 points]: Basic concepts

(single choice just has one solution, while multiple choice has at least two solutions. In these choice questions and true or false questions, you *do not* need to justify your answer.)

(a) [2 points] The least squares learning problem has a unique solution (when data matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d}$ has full column rank) or has no solution (when $n < d$). (true or false)

Solution: False.

(b) [3 points] Is perceptron learning algorithm guaranteed to stop in a certain number of iterations? Briefly justify your answer.

Solution: No. For non-linearly separable data, it will never stop.

(c) [3 points] Describe the key difference between classification, regression, and clustering.

Solution: Classification and regression are supervised learning, in which they have labels. Clustering is unsupervised and does not have labels. Classification uses categorical labels, while regression has continuous labels.

(d) [2 points] Learning is possible in general. (true or false)

Solution: False.

(e) [2 points] Robust linear regression is more robust to outliers than least square. (true or false)

Solution: True.

(f) [3 points] Choose the true statement. (single choice)

(1) Linear model is linear in data sample $\boldsymbol{x}$.
(2) Linear model means that the learning problem is a linear optimization problem.
(3) None of them is true.

Solution: (3).

(g) [3 points] Least squares is a maximum likelihood estimator. (true or false)

Solution: False. (need Gaussian noise assumption)

(h) [3 points] To learn the underlying target function g, we can fit perfectly the training data $\{(\boldsymbol{x}_i, y_i)\}$. Then, the target g is learned. (true or false)

Solution: False.

(i) [3 points] Consider a hypothesis space $\mathcal{H}$. Suppose the growth function $\mathcal{G}_{\mathcal{H}}(2) = 4, \mathcal{G}_{\mathcal{H}}(3) = 8, \mathcal{G}_{\mathcal{H}}(4) = 14, \mathcal{G}_{\mathcal{H}}(5) = 22, \mathcal{G}_{\mathcal{H}}(6) = 32$. What is the the VC dimension d_{VC} of $\mathcal{H}$?

Solution: The growth function $\mathcal{G}_{\mathcal{H}}(n) = 2^n$ breaks when $n = 4$, so we have $d_{\text{VC}} = 3$.

(j) [3 points] From VC analysis,

$i.$ if we have more training data points (larger n), then will the generalization error be larger or smaller?
$ii.$ if we choose a less complex $\mathcal{H}$, then will the generalization error be larger or smaller?

Solution: $i.$ Smaller. $ii.$ Smaller.

(k) [3 points] Suppose the number of training samples n is fixed. Describe the process of making the out-of-sample error small from a VC analysis point of view.

Solution: Choose an appropriate hypothesis space $\mathcal{H}$, then design an optimization algorithm to select $f_{\widehat{\theta}} \in \mathcal{H}$. In this way, $f_{\widehat{\theta}}$ will have simultaneously small generalization error and in-sample error. Through VC analysis, this leads to a small out-of-sample error.

(l) [3 points] *i*. Write down the logistic regression model (binary case). *ii*. Suppose that you have learned a model $\widehat{\boldsymbol{\theta}}$ for (binary) logistic regression. Now, we have a test data $\boldsymbol{x}$. Describe how can you classify $\boldsymbol{x}$ based on the learned model.

Solution: *i*.

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{1}{1 + \exp\left(-y \cdot \boldsymbol{\theta}^\top \boldsymbol{x}\right)}.$$

ii. We can plug $\boldsymbol{x}$ in the learned model $f_{\widehat{\boldsymbol{\theta}}}(\boldsymbol{x}) = \frac{1}{1+\exp\left(-y \cdot \boldsymbol{\theta}^\top \boldsymbol{x}\right)}$ with $y = +1$ and $y = -1$. Then classify $\boldsymbol{x}$ for that class if $f_{\widehat{\boldsymbol{\theta}}}(\boldsymbol{x}) > \frac{1}{2}$. There is also a linear way, i.e., compute $\mathrm{sign}(\boldsymbol{\theta}^\top \boldsymbol{x})$. Then, classify $\boldsymbol{x}$ as $+1$ if the result is $+1$ and vice versa.

(m) [4 points] Choose the true statements about optimization algorithms. (multiple choice)

(1) GD utilizes first-order linear approximation of the learning problem.
(2) GD can be used to learn a soft-margin SVM classifier.
(3) Nesterov's accelerated GD (AGD) can be viewed as GD with a special step size.
(4) For large scale learning problem, SGD converges faster than GD.
(5) In one epoch, random reshuffling will use all the training data points for updating.
(6) For logistic regression learning problem, AGD converges slower than GD.
(7) SGD needs diminishing step sizes for convergence.

Solution: (1), (4) , (5), (7).

(n) [3 points] List three components that may cause overfitting.

Solution: Noise, model complexity, number of training data n, target complexity (list three of them).

(o) [3 points] Roughly describe the idea of validation.

Solution: Split the training dataset into a new training $\mathcal{S}_{\text{train}}$ and a validation dataset $\mathcal{S}_{\text{val}}$. Use the validation set $\mathcal{S}_{\text{val}}$ to compute validation error Er_{val}. With appropriate setting, Er_{val} should estimate well the Er_{out} of f' learned by using $\mathcal{S}_{\text{train}}$. Then, we adjust hyperparameters based the this estimated Er_{out} to avoid overfitting (that gives a small estimated Er_{out}).

(p) [3 points] Choose the wrong statement about regularization. (single choice)

(1) Regularization is to estimate Er_{out}.
(2) Regularization helps reducing overfitting.
(3) ℓ_2-regularized least square almost always admits a unique solution with properly chosen regularization parameter λ, even when the number of training data is less than the feature dimension.
(4) Proximal GD (PGD) has closed-form update when it is used to solve ℓ_1-regularized logistic.

Solution: (1).

(q) [3 points] (true or false). AdamW is to apply Adam to the following regularized problem:

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta}) + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2, \quad \lambda > 0.$$

Solution: False. AdamW uses decoupled weight decay, which is to apply GD to the regularizer $\frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$.

(r) [3 points] Choose the wrong statement.

(1) Neural network (NN) is a nonlinear model.
(2) NN is to perform linear regression or linear classification in the final layer using the learned representation.
(3) Backpropagation (BP) has a forward pass and backward pass.
(4) In BP, we do not need to store anything in the forward pass.

Solution: (4).

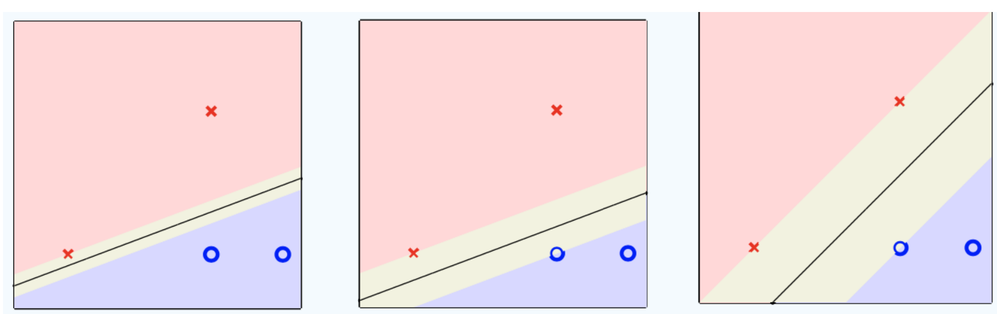
(s) [4 points] Choose the wrong statements. (multiple choice)

(1) Attention mechanism is used to capture semantic structure in a sentence.
(2) Determining $\boldsymbol{y}_i$ of i-th token will only use input embedding $\boldsymbol{x}_i$ in attention.
(3) Self-attention uses inner product between token's embedding to measure similarity.
(4) Multi-head attention uses the same query, key, and value matrices among different heads.
(5) Layer normalization does not have trainable parameters.
(6) Language models usually use transformer to model conditional probability, and then predict the next token using autoregressive modeling.

Solution: (2), (4), (5).

Question 2 [9 points]: SVM and Kernel Methods

(a) [2 points] Which one of the following classifiers is most likely learned by SVM? Briefly justify the reason.



Solution: The right most one since it has the largest margin.

(b) [3 points] Briefly describe why SVM classifier is robust for binary classification.

Solution: We often have *noise* in measurements in practice (1pts). A SVM classifier has the largest margin, so it will be more robust to noise; the test data can be viewed as training data with error; hence, SVM classifier is the optimal one in total (2pts).

Note: Another point of view is from regularization: SVM classifier is a regularized perceptron classifier. This leads to the robust feature of SVM. This can also earn (2pts)

(c) [4 points] Derive the learning problem of kernelized soft-margin SVM with kernel function κ, by replacing $\boldsymbol{\theta} = \sum_{j=1}^{n} \alpha_j \boldsymbol{x}_j$, where $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$ are the training data.

Solution:

$$
\begin{aligned}
\widehat{\boldsymbol{\alpha}} &= \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \ \frac{1}{n}\sum_{i=1}^{n} \max\left(0, 1 - y_i\left(\sum_{j=1}^{n} \alpha_j \boldsymbol{x}_j^\top \boldsymbol{x}_i + b\right)\right) + \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i\alpha_j \boldsymbol{x}_i^\top \boldsymbol{x}_j \quad \text{(2pts)} \\
&\overset{\text{kernelize}}{=} \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \ \frac{1}{n}\sum_{i=1}^{n} \max\left(0, 1 - y_i\left(\sum_{j=1}^{n} \alpha_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) + b\right)\right) + \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i\alpha_j \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) \quad \text{(2pts)} \\
&= \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \ \frac{1}{n}\sum_{i=1}^{n} \max\left(0, 1 - y_i\left(\boldsymbol{K}_i^\top \boldsymbol{\alpha} + b\right)\right) + \lambda \boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}
\end{aligned}
$$

where $\boldsymbol{K}(i,j) = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is the kernel matrix and $\boldsymbol{K}_i^\top$ is its i-th row.

Question 3 [10 points]: Neural Network

An L layer deep forward neural network is

$$f(\boldsymbol{x}) = h(\boldsymbol{W}_L\sigma(\boldsymbol{W}_{L-1}\cdots\sigma(\boldsymbol{W}_2\sigma(\boldsymbol{W}_1\boldsymbol{x})))),$$

where $\boldsymbol{x} \in \mathbb{R}^{d_0}$ is the input, $\boldsymbol{W}_l \in \mathbb{R}^{d_l \times d_{l-1}}$ is the weight of $l-$th layer, $\sigma(\cdot)$ and $h(\cdot)$ are the activation functions.

(a) [2 points] Suppose the input data $\boldsymbol{x}$ is of dimension 10 and the output label y is of dimension 1. Consider the 3 hidden layer neural network (with no bias terms) with neurons of $\{12, 24, 8\}$. Calculate the total parameter number for the neural network.

Solution: We have in total $10 \times 12 + 12 \times 24 + 24 \times 8 + 8 = 608$ number of parameters.

(b) [4 points] Consider the scalar input $x \in \mathbb{R}$. Let $h(\cdot)$ be the identity function and $\sigma(\cdot)$ be the Relu activation function:

$$h(x) = x, \quad \sigma(x) = \max\{x, 0\}.$$

Design a neural network $f(x) = x$. Specify the weight of each layer.
(Hint: $x = \max\{x, 0\} - \max\{-x, 0\}$)

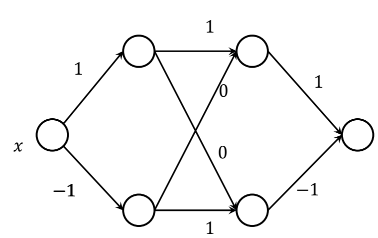


Figure 1: Neural network design for problem (b).

Solution: Observe that $x = \max\{x, 0\} - \max\{-x, 0\}$ (1pts). We can construct the following 2 hidden layer neural network. The weights are

$$\boldsymbol{W}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{W}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{W}_3 = \begin{bmatrix} 1 - 1 \end{bmatrix}. \quad \text{(3pts)}$$

(c) [4 points] Consider the input $\boldsymbol{x} = (x_1, x_2) \in \mathbb{R}^2$. Let $h(\cdot)$ be the identity function and $\sigma(\cdot)$ be the piece-wise ReLU activation function:

$$h(\boldsymbol{x}) = \boldsymbol{x}, \quad \sigma(\boldsymbol{x}) = \begin{bmatrix} \max\{x_1, 0\} \\ \max\{x_2, 0\} \\ \vdots \\ \max\{x_{d_0}, 0\} \end{bmatrix}.$$

Design a neural network $f(\boldsymbol{x}) = \max\{x_1, x_2\}$. Specify the weight of each layer.

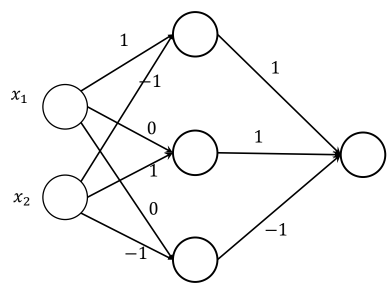


Figure 2: Neural network design for problem (c).

Solution: Observe that

$$\begin{aligned} \max\{x_1, x_2\} &= \max\{x_1 - x_2, 0\} + x_2 \\ &= \max\{x_1 - x_2, 0\} + \max\{x_2, 0\} - \max\{-x_2, 0\} \quad \text{(2pts)} \end{aligned}$$

We can construct a 1 hidden layer neural network. The weights are

$$\boldsymbol{W}_1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{W}_2 = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}. \quad \text{(2pts)}$$

Question 4 [10 points]: PCA

Consider the following form of PCA:

$$\underset{\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}}{\text{maximize}} \sum_{i=1}^{n} \|\boldsymbol{A}^\top \boldsymbol{x}_i\|_2^2,$$

where $\boldsymbol{x}_i \in \mathbb{R}^d$ and $\boldsymbol{A} \in \mathbb{R}^{d \times k}$.

(a) [3 points] How to compute the solution $\boldsymbol{A}$?

Solution: We first form the data matrix $\boldsymbol{X} \in \mathbb{R}^{d\times n}$ with each column being $\boldsymbol{x}_i$ (1pts). Then, we compute the SVD of $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$. Then, $\boldsymbol{A} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k]$ (2pts).

(b) [3 points] After computing $\boldsymbol{A}$, how to compute the principle component matrix $\boldsymbol{\Theta}$? What is the dimension of $\boldsymbol{\Theta}$?

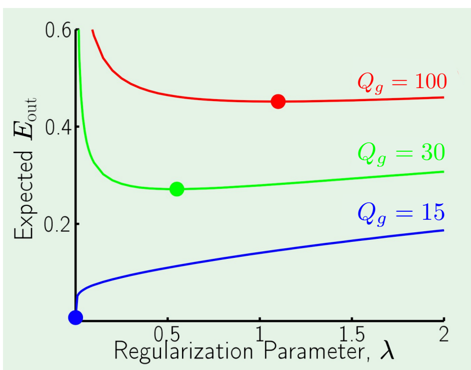
Solution: $\boldsymbol{\Theta} = \boldsymbol{A}^\top \boldsymbol{X}$ (2pts), which is of dimension $k \times n$ (1pts).

(c) [4 points] Suppose we increase k, will the error $\sum_{i=1}^n \|\boldsymbol{x}_i - \boldsymbol{A}\boldsymbol{\theta}_i\|^2$ be smaller or larger? Here, $\boldsymbol{A}$ is the solution in Part (a) and $\boldsymbol{\theta}_i$ is the i-th column of $\boldsymbol{\Theta}$ in Part (b). Briefly justify the answer.

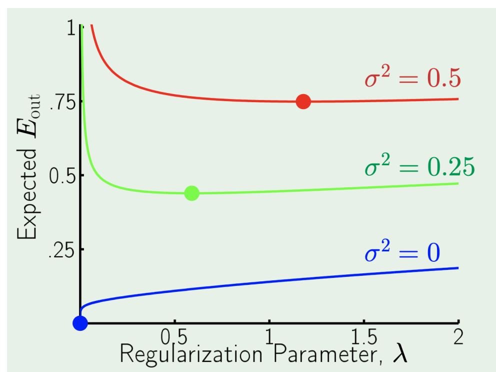
Solution: Will decrease (2pts). With larger k, we will have a more accurate decomposition, hence the reconstruction error $\sum_{i=1}^n \|\boldsymbol{x}_i - \boldsymbol{A}\boldsymbol{\theta}_i\|^2$ of PCA will be smaller (note that these two formulations of PCA are equivalent) (2pts).

Question 5 [15 points]: Overfitting

We will discuss overfitting, validation, and regularization (wight decay) in this question.



(a) Fixed noise level: $\sigma^2 = 0$.



(b) Fixed target model complexity $Q_g = 15$.

Figure 3: Evolution of Er_{out} with respect to the regularization parameter λ, where the optimal λ with lowest Er_{out} in each curve is dotted. In the figure, Q_g is the target model complexity (i.e., the underlying target model g's complexity), and σ^2 is the noise level in the training data. In both Figure 1 (a) and Figure 1 (b), we use a model f with complexity $Q_f = 15$ to fit the training data using weight decay regularization with a regularization parameter $\lambda \geq 0$.

(a) [6 points] Consider Figure 3 (a).
(1) Explain the reason why the blue curve ($Q_g = 15$) increases Er_{out} with increasing λ.
(2) Explain the reason why the optimal λ increases with increasing target complexity Q_g.

Solution: (1) In the blue curve, stochastic noise level $\sigma^2 = 0$ and $Q_f = Q_g$, we have the good fitting setting. Hence, $\lambda = 0$ is optimal since there is no overfitting (1pts). With increasing λ, it penalizes the ability of f to fit the data, thus underfitting (higher Er_{out}). (2pts)

(2) We have mismatch between underlying target model and the complexity of f. Hence, higher the Q_g more *deterministic noise* we have, as the mismatch can be viewed as deterministic noise (2pts). Thus, larger mismatch (higher Q_g) needs larger regularization (higher λ) (1pts).

(b) [5 points] Consider Figure 3 (b).
   (1) In the blue curve, is the model underfit, goodfit, or overfit when $\lambda = 1$?
   (2) Explain the reason why the green (or red) curve first decrease to the lowest $\mathrm{Er}_{\mathrm{out}}$, then increase $\mathrm{Er}_{\mathrm{out}}$.

Solution: (1) The interpretation is the same as the blue curve in Figure 3 (a). With $\lambda > 0$, we penalize the fitting ability of f in the non-overfitting case, it is thus underfit with $\lambda = 1$. (2pts)
(2) Since we have noise, too small lambda leads to overfit to the stochastic noise in the data and hence large $\mathrm{Er}_{\mathrm{out}}$ (1pts). With increase of λ, we are near goodfit, which gives lowest $\mathrm{Er}_{\mathrm{out}}$ (1pts), Then, if we continue to increase, we are in the underfit regime hence increasing $\mathrm{Er}_{\mathrm{out}}$ again (1pts).

(c) [4 points] Suppose that we are in the regime of binary linear classification and the definitions of $\mathrm{Er}_{\mathrm{in}}$ and $\mathrm{Er}_{\mathrm{out}}$ are both using binary error. In the context of validation, show the following inequality appeared in our lecture slides, where k is the validation dataset size:

$$\mathrm{Er}_{\mathrm{out}}(f') \leq \mathrm{Er}_{\mathrm{val}}(f') + \mathcal{O}\left(\frac{\log(1/\delta)}{\sqrt{k}}\right),$$

which holds true with probability at least $1 - \delta$. Here, f' is trained on the $n - k$ new training set.

Solution: The key of this result is that we only need to derive the bound for a single hypothesis f', no union bounds needed (2pts). The proof (answer) will be exactly the same as that of the generalization bound for a fixed f, i.e., repeat the same thing from pp. 8 of slides6; and let $\delta = e^{-2nt^2}$. Listing the proof worth (2pts).